

Doc. 316 – FFGFT and the Riemann Zeta Function

Reach and Limit: Two Identities, Four Exclusions

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Preliminary: the answer in short

Result

The Riemann hypothesis is an **arithmetic** matter. The torus and ξ contribute nothing to its solution. The result of this document is a *negative* one – three obvious geometric approaches are excluded with reasons given.

The blanket formulation would be too coarse in one respect, hence the refinement.

What remains purely a matter of numbers. Where the zeros lie, how they stand relative to one another, how one grasps them individually – that is carried by arithmetic, by the primes. Three geometric approaches were tested and all are negative: no compact geometry can carry the zeros as its spectrum (Section .3, at theorem level), ξ does not shift the critical line (Section .2), and the ξ ladder discretises nothing (Section .5, out-of-sample test negative).

What is nevertheless not merely a matter of numbers. Two things. First, the Riemann zeta is a *factor* in the spectrum of the $\mathbb{Z}^4$ torus – an identity, not a picture (Section .2). The Riemann hypothesis is thereby a statement about a geometric object of the framework; only one that the geometry cannot answer. Second, the form of the zero density says *where* the question belongs: $N(T)$ grows like $T \ln T$, and no Laplacian on any compact manifold can deliver that – it requires a dilation structure, hence the unrolled domain. The geometry indicates the address without supplying the answer.

The striking thing is precisely that both occur together: the numbers demonstrably sit in the torus spectrum, but the torus cannot generate them.

The yield. Three dead ends are closed, with reasons rather than conjecture. In addition, one of the author's own faulty constructions is documented: a cell argument that initially looked load-bearing turned out on elaboration to be a restatement of $\hbar = 1$ (Section .4) and was withdrawn.

.1 Starting point and claim

[Q] The corpus mentions the Riemann hypothesis in exactly one place: Doc. 024 notes, in an AI/network context, that the "number space" has dimensions "modulated by the Riemann hypothesis and ζ functions". That is a conceptual marginal remark without an epistemic marker and without derivation. A worked-out connection does not exist in the corpus.

[K] At the same time Doc. 314 contains material that touches the ζ function directly: Epstein zeta, Casimir point, heat kernel on T^4 . This document examines systematically what follows.

Claim. It proves nothing about the Riemann hypothesis and does not claim to. Three things are clarified: which connections hold as identities **[K]**, which path carries constructively **[B]**, and which obvious paths are *demonstrably* not passable **[X]**. The exclusion is the most valuable part – it saves work on dead ends.

.2 Two findings that hold as identities

The Riemann zeta is a factor of the torus spectral zeta

[K] Doc. 314 carries the closed form of the $\mathbb{Z}^4$ spectral zeta (verified threefold there; confirmed here against the direct sum with analytic tail correction to 10^{-7} through 10^{-12}):

$$Z_{\mathbb{Z}^4}(s) = 8(1 - 4^{1-s})\zeta(s)\zeta(s-1).$$

This is not an analogy but an identity. The zeros of the torus spectral zeta therefore have three sources:

Source	Location of the zeros
$\zeta(s) = 0$	$\Re(s) = 1/2$ (non-trivial – the RH)
$\zeta(s-1) = 0$	$\Re(s) = 3/2$ (shifted by 1)
$1 - 4^{1-s} = 0$	$s = 1 + 2\pi i k / \ln 4$ (lattice factor)

On this reading the Riemann hypothesis is a statement about the spectrum of the $\mathbb{Z}^4$ torus – about a geometric object of the framework, not merely about a number series.

The Casimir point lies symmetric to the critical line

[K] Doc. 314 computes the ζ -regularised vacuum energy $E = \pi\zeta_M(-1/2)$ and finds the order reversal: D4 minimises the Epstein zeta for $s > 0$, at $s = 0$ all lattices are equal, and at $s = -1/2$ the $\mathbb{Z}^4$ torus lies lower (-0.9321 against -0.8686).

The Casimir point $s = -1/2$ has, under the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$, the mirror point $s = 3/2$. The axis of symmetry:

$$\frac{-1/2 + 3/2}{2} = \frac{1}{2} \quad - \text{ exactly the critical line.}$$

The physically occupied interval $[-1/2, 3/2]$ has the critical line as its midpoint: consistency between physics and arithmetic.

[X] No proof follows from this – the critical line is fixed by the functional equation anyway. Nor can ξ shift it: a perturbation of order $\xi \sim 10^{-4}$ (even with the RG amplifier 100ξ) does not reach an axis fixed by a functional identity.

.3 The decisive obstruction: Weyl against Riemann-von Mangoldt

The obvious move would be to seek an operator whose eigenvalues are the zeros (Hilbert-Pólya programme), and to deform $T^4/\mathbb{Z}_3$ suitably for that purpose. **This entire branch is excluded – at theorem level, not as intuition.**

[Q] Weyl's law. For every compact Riemannian manifold of any dimension d , asymptotically $N(\lambda) \sim C_d \text{Vol} \lambda^{d/2}$ – a pure power. Curvature, boundary and orbifold points change only lower orders, not the leading term.

[Q] Riemann-von Mangoldt (proved).

$$N(T) = \frac{T}{2\pi} \ln \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + O(1/T).$$

[K] The comparison via the local exponent $d \ln N / d \ln T$:

T	$N(T)$	local exponent $d/2$
10^3	$6.48 \cdot 10^2$	1.2457
10^5	$1.38 \cdot 10^5$	1.1153
10^8	$2.48 \cdot 10^8$	1.0642
10^{12}	$3.95 \cdot 10^{12}$	1.0403
10^{20}	—	1.0233

The exponent is $1 + 1/\ln(T/2\pi)$ – it falls monotonically towards 1 but is *never constant*. Range over 17 decades: 0.22. The best power fit in the window $[10^3, 10^6]$ deviates systematically by up to 11 %.

Exclusion at theorem level

[X] There is no compact Riemannian manifold – of whatever dimension, curvature, topology or orbifold structure – whose Laplace spectrum is the Riemann zeros. Reason: $T \ln T$ is not a power $T^{d/2}$.

Consequence. The branch “deform $T^4/\mathbb{Z}_3$ suitably” is settled completely, not only in the flat case. The extra $\ln T$ is the signature of a **scale degree of freedom**: a dilation direction with logarithmically growing volume. That is not a Laplacian on a manifold.

Withdrawn intermediate diagnosis. In a first pass it was argued that $\log(p)$ geodesics require negative curvature. That is wrong: what generates log lengths is a dilation structure (multiplicative group $\mathbb{R}_+^*$), not curvature; the Berry-Keating operator $H = xp$ is the generator of dilation, not of a hyperbolic flow. The Weyl obstruction is the correct and considerably stronger statement, and it renders the curvature question moot.

.4 The constructive link – and why it does not carry

If the zero density requires a scale degree of freedom, the responsible domain is the **unrolled** one (Doc. 315) – multiplicative, logarithmic, with dilation as its natural operation.

The approach

[Q] Berry-Keating (1999). The semiclassical count of the dilation generator $H = (xp + px)/2$ in a phase space with a minimal cell:

$$N(E) = \frac{1}{2\pi} [E \ln(E/(l_x l_p)) - E + l_x l_p].$$

With $l_x l_p = 2\pi$ this becomes exactly the Riemann-von Mangoldt formula (verified: difference constant $0.125 = 1 - 7/8$, a pure boundary term, over $T = 15 \dots 5000$).

The obvious chain would then be: Doc. 314 Ch. D1 carries $\lambda_4 \cdot m = 2\pi$ as an exact position relation; in the unrolled domain the quantity conjugate to scale is the dilation momentum; hence FFGFT supplies the cell condition.

Why the chain collapses

[X] Stage 1 – the cell condition is empty of content. $l_x l_p = 2\pi\hbar$ is, in natural units, the Planck cell, and follows from the canonical quantisation of every conjugate pair. That $\lambda_4 m = 2\pi$ has the same form is no independent finding – the de Broglie relation *is* the statement $\hbar = 1$. Moreover the count is logarithmically insensitive to the cell (a factor 2 wrong $\Rightarrow N$ changes by $\sim 1\%$); a “hit” would not have been a test in any case.

[X] Stage 2 – the actual content fails quantitatively. Berry-Keating requires something sharper than the cell: two *independent* cutoffs $x > l_x$ and $p > l_p$ whose product is the Planck cell. The FFGFT double cutoff makes this testable:

Quantity	Value
UV: $L_0 = \xi l_p$	$2.155 \cdot 10^{-39} \text{ m}$
IR: $L^* = 4\tilde{\lambda}_e/\xi^{10}$	$8.698 \cdot 10^{26} \text{ m}$
Ratio L^*/L_0	$4.04 \cdot 10^{65}$
required	$2\pi \approx 6.28$
Shortfall	65 decades

[X] Stage 3 – a finite window against infinitely many zeros. $\ln(L^*/L_0) = 151.06$, that is 16.93ξ ladder rungs; the Berry-Keating count within the window gives ~ 1816 modes. A finite scale window carries finitely many modes; the zeros are countably infinite. Even if stages 1 and 2 held, at most an initial segment would result – and an initial segment is not a spectral identity.

Marginal observation, explicitly not booked: 16.93 lies close to 17 . Both cutoffs carry their own uncertainties – the prefactor 4 is open per R73, l_p depends on G . Without an error budget this is not assessable under P35.

What remains

To be withdrawn is the formulation “the zero density follows from the FFGFT cell condition”. It suggests a finding where a triviality stands.

[B] A structural statement remains, and it is not little: that the counting function has the form $(T/2\pi) \ln(T/2\pi)$ at all requires a **dilation generator**. A Laplacian on a compact manifold cannot deliver it (Section .3). The substantive statement is the *identification of the structure* – the unrolled domain carries the dilation – not the numerical value 2π .

.5 The missing boundary condition – test negative

$H = xp$ has *continuous* spectrum. The counting function fixes the mean density, not the individual zeros. What is missing is the boundary condition that turns the continuum into a discrete spectrum.

The natural candidate. The ξ ladder generates a discrete subgroup $\Gamma_\xi = \{\xi^n\} < \mathbb{R}_+^*$; the quotient $\mathbb{R}_+^*/\Gamma_\xi$ is a *circle* of circumference $L = \ln(1/\xi) = 8.9227$. Admissible dilation frequencies would then be multiples of $2\pi/L = 0.7042$.

Test discipline

[K] Proximity arguments require three controls (P35): a null distribution over random periods in the *same* range; artefact control, because periods larger than the data range automatically produce small scores; and an out-of-sample check with look-elsewhere correction.

The artefact control is not theoretical: a first run reported an apparent signal at period $2\pi/|\ln(74/75)| = 468$. Since all zeros lie < 102 , all quotients round to 0 – a pure artefact of a null distribution that did not cover the range. Documented because exactly this error type recurs with ξ proximity arguments.

Result

[K] Candidates within the admissible range (null distribution: mean 0.2502):

Candidate	Period	Score	p value
$2\pi/\ln(1/\xi)$	0.7042	0.2812	0.92
$2\pi/\ln 75$	1.4553	0.2043	0.005
$\ln(1/\xi)$	8.9227	0.2891	0.94
2π	6.2832	0.2366	0.18

The borderline candidate $2\pi/\ln 75$ was checked out of sample (zeros 31–50): score 0.2686, $p = 0.81$; look-elsewhere corrected $p = 1.0$.

Boundary condition

[X] The ξ ladder does not supply the missing boundary condition. No ξ -derived period candidate survives the out-of-sample check.

Why this was structurally to be expected

[K] A single-period compactification produces an *equidistant* lattice. The zeros, however, show GUE statistics with level repulsion (Montgomery-Odlyzko): mean normalised spacing 0.9815 (should be ~ 1), fraction of spacings < 0.3 equal to 0.000 – against ~ 0.04 for GUE and ~ 0.26 for a lattice-like spectrum. A circle cannot deliver GUE in principle.

The missing boundary condition is carried by arithmetic (the primes), not by geometry.

.6 Balance

Statement	Status
ζ is a factor of the torus spectral zeta	[K] verified
Casimir interval has critical line as axis	[K]
Zero density requires dilation generator	[B] structure
Cell condition as an independent finding	[X] only $\hbar = 1$
Double cutoff as BK truncation	[X] 65 decades
Compact manifold with zero spectrum	[X] Weyl
ξ ladder discretises the zeros	[X] test negative
ξ shifts the critical line	[X] functionally fixed
FFGFT solves the Riemann hypothesis	[X]

The most honest sentence. Of the three connections examined, two hold as identities **[K]** – but they are consistency, not a statement about the location of the zeros. The third, the constructive one, collapsed on elaboration: what looked like a derivation is $\hbar = 1$. What remains is a structural statement: the zero density requires a dilation generator, and that lives in the unrolled domain.

What a viable approach would have to achieve: a non-trivial boundary condition on the unrolled scale direction that (a) carries infinitely many modes – hence is not a finite cutoff window – and (b) produces GUE statistics rather than equidistance. Both requirements point in the same direction: the structure must be arithmetic, not geometric.

.7 Open items

- **Z-1:** Does the ξ ladder carry information about the Euler product? $\ln(1/\xi) = 2 \ln 2 + \ln 3 + 4 \ln 5$ decomposes exactly into the prime factors of 30000. Whether that is more than the trivial prime factorisation: open, no test formulated so far.
- **Z-2:** The lattice factor $(1 - 4^{1-s})$ yields zeros at $s = 1 + 2\pi i k / \ln 4$. Does this ladder have a corpus meaning (D4? the factor 4 from $L^* = 4\tilde{\lambda}_e/\xi^{10}$, R73)? Not yet examined.
- **Z-3:** The heat-kernel finding of Doc. 314 (D4/ $\mathbb{Z}^4$ difference exponentially small, $1.8 \cdot 10^{-13}$ at $t = 0.1$) has a counterpart in ζ theory (theta-series fine structure). Whether a test surface follows: open.
- **Not recommended:** further search for ξ periods in the zeros. Section .5 excludes this structurally via the GUE argument, not merely numerically.

.8 Verification

All numbers are reproducible by five scripts (standard library only, target values as assertions, seed 20780458):

- `z1_weyl_obstruktion.py` – Weyl’s law against Riemann-von Mangoldt, local exponent, power fit.
- `z2_zellbedingung.py` – Berry-Keating count, equivalence to the Riemann form, cross-check against Weyl.
- `z3_randbedingung_test.py` – ξ ladder test with artefact control, null distribution, out-of-sample, GUE control.
- `z4_epstein_casimir.py` – factorisation against the direct sum, Casimir symmetry point.

- `z5_zellbedingung_kritik.py` – collapse of the cell argument in three stages.

Inputs: $\xi = 4/30000$ exact; $\ell_p = 1.616255 \cdot 10^{-35}$ m, $\bar{\lambda}_e = 3.8615926796 \cdot 10^{-13}$ m (CODATA 2022); zero values (first 50) from the literature.